

Digital Twin Framework for Personalized Serious Games-Based Therapy

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Abstract—Patients with disabilities usually require repetitive and sometimes painful therapy to regain functionality. Serious games are used as a tool to complement therapist’s interventions aiming to promote engagement and motivation. Smart healthcare provides tools to build smart environments using technological advances such as wearable, mobile, and Internet of Things devices to capture data from user’s interactions to obtain knowledge. A digital twin of the patient is a representative virtual replica of the physical patient allowing status assessment, application of scenarios, simulation, and prediction. A framework model for a digital twin of the patient for personalized serious game-based therapy is presented.

Index Terms—Digital Twin, Framework, Serious Games, Personalized Therapy

I. INTRODUCTION

According to the World Health Organization (WHO), 16% of the world population experience significant disability [1]. People with disabilities present conditions limiting their skills in some way, with the impairments being chronic or temporary, significantly degrading their health and well-being and increasing the risk of facing health inequities and developing other conditions. Disabilities result from the interaction between health conditions and impairments from pathologies such as dementia, blindness, spinal cord injury (SCI), cerebral palsy (CP), Parkinson’s disease (PD), Alzheimer’s disease, stroke, speech sound disorders (SSD), autism spectrum disorder (ASD), attention deficit hyperactivity disorder (ADHD). Patients with disabilities often need personalized therapy exercises to maintain or regain functionality. Each patient has distinct levels of pathology, usually with a combination of multiple conditions. It is important to consider the individual patient’s several dimensions. Children and older adults, as patients with disabilities, have distinct requirements than regular adults. Therapy for these patients is usually repetitive and sometimes painful to be able to get some valid therapeutic results.

The characteristics of the focused pathologies, required therapies, and population make it difficult for patients to keep

their therapy routines. Techniques such as gamification and serious games (SG) are commonly used to help therapists and patients overcome this relevant constraint. Gamification refers to the use of game elements in other contexts. An SG is a game with a purpose where the main goal is not pure entertainment. SGs are used for motivation and engagement to help patients maintain their therapy routines [2]. SGs promote therapeutic exercises as game actions.

Smart healthcare (SH) is a growing field in medicine, using technological advances such as wearable, mobile and Internet of things (IoT) devices to provide tools for communication and access to information [3]. Smart environments use sensors and IoT devices to capture data from user’s interactions. When focusing patients with disabilities, the use of natural user interface (NUI) sensors are a relevant solution to overcome the difficulty in using common interfaces. Virtual reality (VR) and augmented reality (AR) can be used to provide immersive and extended environments for users. Collected data are usually stored in electronic health records (EHR) platforms. These data includes large volumes of heterogeneous multimodal data (e.g., demographic, biometric, clinical analysis and reports, exercise results, audio, video, images, text, and sensor data). The vast amounts of collected data quickly becomes unmanageable for interpretation by domain experts. Artificial intelligence (AI) and machine learning (ML) algorithms are used to process this big data to make it more manageable and interpretable. These technological advances contribute to the development of representative models of patients.

This research aims at providing a model of a digital twin (DT) of the patient for personalized SG-based therapy. It is expected that this DT allows application of scenarios, simulation and prediction as a tool to achieve the best therapeutic results.

II. RELATED WORK

A DT consists of a virtual model of a physical entity and the data connection between them [4]. A DT can be

characterized by a series of distinct features including the physical and virtual twins, physical and virtual environments, state, realisation, metrology, twinning methods and rate, bidirectional connection and data flow, and the physical and virtual processes [4]. State is defined by the current value of all parameters, and refers to the current condition of both virtual and physical twins (e.g., operational, mechanical, state of disease). Enabling technologies for DT include IoT, mobile communications (5G/6G), VR/AR, 3D modeling, AI, distributed computing, Transfer Learning (TL), and sensors allowing seamless monitoring, analysis, evaluation and prediction [5]. The DT concept has been applied to several fields such as science, engineering, and healthcare. Healthcare applications use DTs for improvement, analysis, and prediction for patients, hospitals, and the pharmaceutical industry [6].

A Human DT (HDT) is a digital replica of the human body or specific organ in the body [7], extending the DT concept, and providing extra tools for personalized treatment and preventive care [8]. The DT concept, when modeling the human body, is not clearly defined mainly due to the extreme complexity of humans that must be modeled digitally [3], so existing applications focus on specific biologic parts and functions. Data collected using sensors and IoT devices are used to feed the DT smart layer that use AI and ML algorithms. Examples of collected data are electroencephalography (EEG) for recording brain signals, electromyography (EMG) for detection of muscular activity in a monitored body area, usually arms or legs, electrocardiography (ECG) for monitoring heart activity, electrodermal activity (EDA) for assessing variation in the electrical characteristics of the skin.

After a literature search we can conclude that not many works focus SG-based DT for therapy. IoT is used as supporting technology for most DT conceptual models or implementations. Wang et al. [9] integrated the concepts of DT and SG in a Minecraft application to promote learning through play for virtual urban exploration and urban planning in the context of the restrictions imposed by COVID-19 pandemic. This research uses the DT concept for modeling the physical urban reality. Sepazgozar et al. explore a DT and web virtual gaming platform for learning in the context of construction [10]. The implemented DT models an excavator machine in the immersive VR space. Students use the DT to learn how to operate the excavator before using a real one. Olszewski et al. developed a SG focusing sustainable development to be used in the context of developing a DT of a smart city [11]. The SG was developed in Unity and is based on the engine "Cities: Skylines", adapted for the target city. The DT considers spatial, economic, social, and legal relations allowing simulation of distinct scenarios of city development. Laamarti et al. propose a framework for DT for health and well-being in smart cities using the ISO/IEEE 11073 standard that provides the X73 Personal Health Device Standards for better interoperability [12].

Barricelli et al. propose an HDT for fitness, composed of a set of DT, with each tracking distinct features of an

athlete, for consecutive days [13]. Features measured include food intake, activity, and sleep, using IoT sensors embedded in wearable devices and applications for manual logging. A fitness framework is used together with algorithms including k-Nearest Neighbours (KNN) imputation for missing values in data, and a KNN classifier and Support Vector Machine (SVM) for prediction. After receiving enough data, the DT can predict the athlete's performance during training. He et al. propose a HDT framework focused on human factors (e.g., ergonomics) that evolves from the Digital Human Modeling (DHM) concept integrating multimodal data and AI for continuous monitoring and assessment, allowing real-time analysis, timely feedback, and bidirectional interactions [14], but not focusing SG-based therapy.

III. DIGITAL TWIN FRAMEWORK PROPOSAL

Previous work allowed conceptualization of a framework and implementation of a web platform (PLAY) [15]. PLAY is a multiuser web-based platform for patients, therapists, and clinics, and providing SG design tools, backend services, and communication functionality using a HTTP REST API, and WebSockets (WS) for bidirectional communication with the external devices and applications. Using these tools it is possible to integrate with IoT devices and external SG and other applications. In most use cases the logic for interfacing with external devices is implemented on the SG applications and data collected is then sent to the PLAY server using API calls. The PLAY system architecture can be seen in Fig. 1.

The PLAY platform is a supporting component for the proposed DT of each patient. Each patient registered in the platform has an individual profile gathering demographic and biometric individual data, registered interaction data, exercise performance data, as well as status assessments provided by therapists. Historical data from every SG execution is kept in the PLAY database, including the obtained results and captured samples. PLAY uses its internal API to allow com-

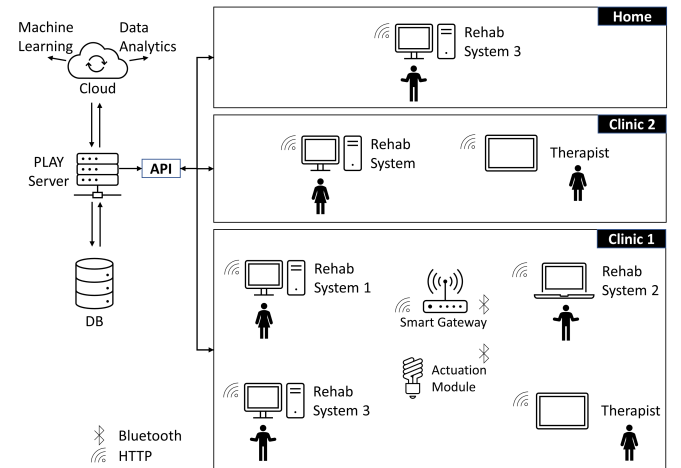


Fig. 1. PLAY system architecture with server, database, and API, showing 3 possible usage scenarios.

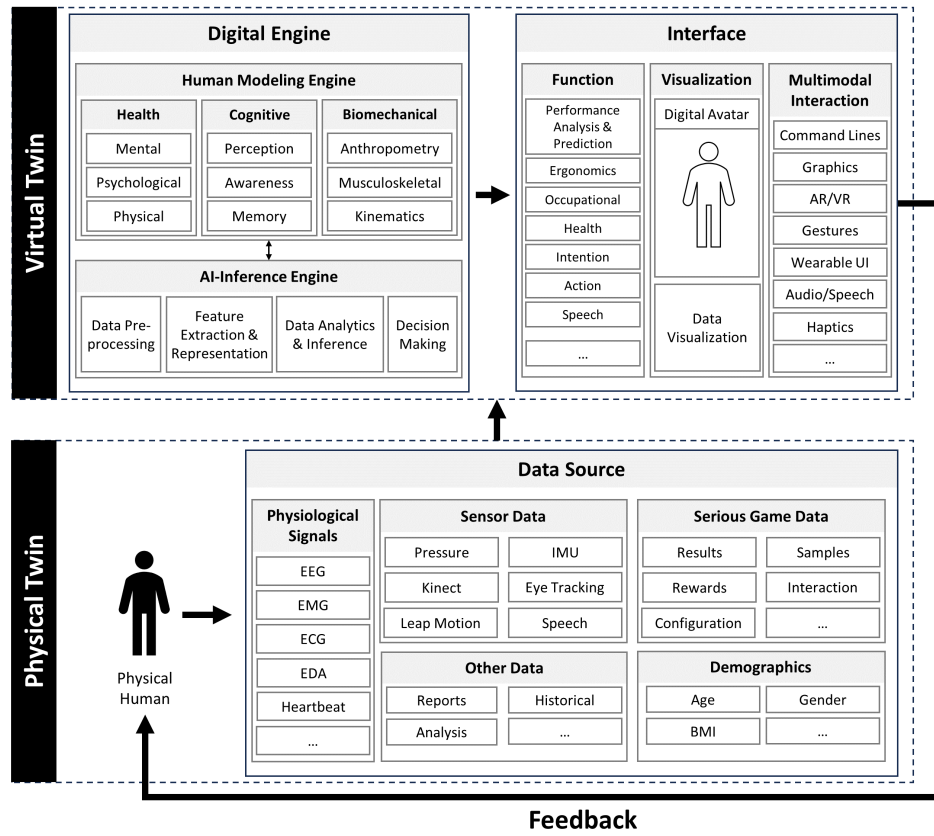


Fig. 2. Digital Twin framework.

munication with external AI solutions for analysis, simulation, and prediction.

The DT system framework includes the physical and virtual twins. The physical twin section is composed by the patient (physical entity) and the data acquisition components (SG with logic for sensor integration) The patient plays one of the SGs integrated with PLAY and focusing specific therapies. According to the focused therapy, each SG implements the logic to integrate with the needed sensors (e.g., IoT, Inertial Measurement Unit (IMU), Brain-computer Interface (BCI), NUI, VR/AR headset) for capturing the relevant signals (e.g., EEG, EMG, ECG, EDA, articulations coordinates). The captured data is sent to the PLAY server using the provided API and recorded in the database, associated to the patient profile. Recorded data is then available to be processed and analysed to be used in AI and ML algorithms. These AI/ML components can be internal or external to the PLAY server. Results obtained from application of the AI components is redirected to PLAY and will update the patient profile status. The API allows the relevant results to be sent directly to the SG in realtime allowing the game flow to be accordingly updated. Therapists can consult all data, results, and obtained information, for decision support, using the PLAY UI tools. Feedback to the patient, closing the DT data flow, can be achieved by automatic adjustment of SG parameters based on the current profile status, and by therapist interactions with the

platform and directly to the patient. The system framework for a personalized DT for each patient can be seen in Fig. 2.

IV. RESULTS

Improvements have been made to the PLAY platform to better conform to the implementation of DT. The REST API and WS communication components have been updated with new functionality.

Several SGs are currently integrated with PLAY, including:

- A SG for upper limb rehabilitation, name JustPhysioKidding (JPK), using the Microsoft Kinect and Leap Motion for capturing articulations features (e.g., coordinates, velocity) during gross and fine motor interactions, and the Shimmer IMU sensor for capturing ECG/EMG/EDA data. User studies have been done. The UI of JPK can be seen in Fig. 3 a).
- A SG for lower limb rehabilitation (PlaymatFun) using the Dance Mat and Wii Balance [16]. Each sample obtained from the Dance Mat is composed by the id of the pressed mat location and the duration of the pressing. Samples obtained using the Wii Balance contains measurements for the weight distribution obtained at each of the 4 balance quadrants, and the calculated coordinates values of the center of mass of the patient. This SG was updated to take in account automatic personalization and adaptation using AI algorithms (clustering) taking in

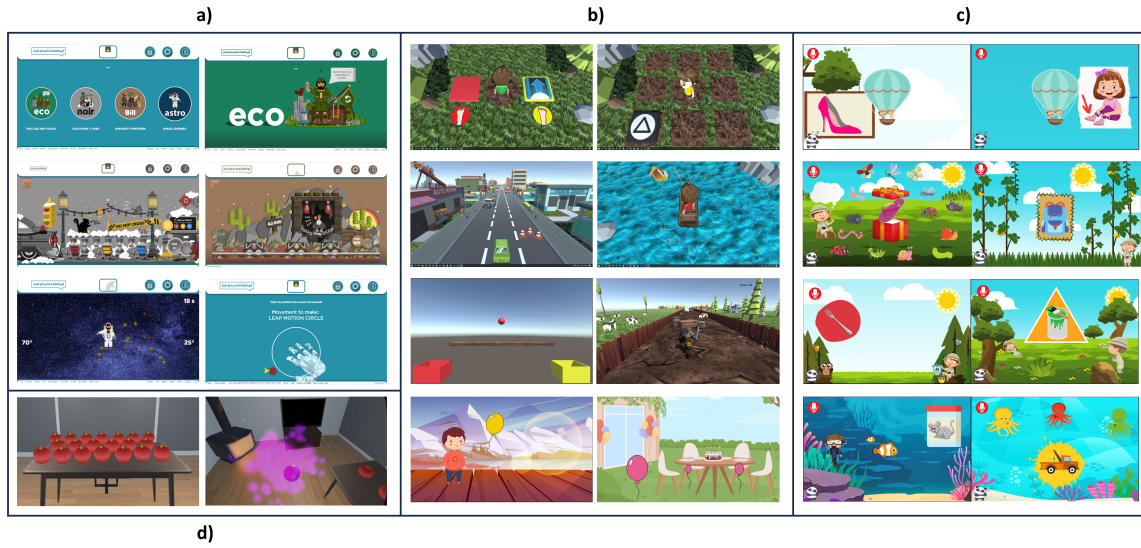


Fig. 3. PLAY SG use cases: a) JPK, b) PlaymatFun, c) Fanima, d) Pikita.

account the specific needs and characteristics of patients. A study with patients at a partner clinic already provided results. The UI of PlaymatFun can be seen in Fig. 3 b).

- A VR-based SG for food phobia (Pikita) capturing user interactions focusing speech and occupational therapy. The game uses the Oculus Quest II and the Microsoft Kinect sensor. PLAY provides a therapist UI for game configuration and realtime control and interaction during the game. The UI of Pikita can be seen in Fig. 3 d).
- A VR-based SG with eye tracking for assessing the cognitive load of patients during interaction tasks. Captured data during tasks (e.g, pupil dilation, saccades) are stored and processed in PLAY. The processed cognitive load signal can be seen in Fig. 4.
- A VR-based SG for PD using the Oculus Quest II and the Microsoft Kinect sensor, to motivate exercise practice following therapy programs, and focusing physiotherapy. A user study has been done.
- A SG for speech therapy (Fanima) allowing a phonetic-phonological assessment test, focusing children with SSD and capturing the audio samples for classification

of speech errors [17]. Fanima uses the therapist's PLAY frontend as a cooperative complement for controlling the flow of the game and for speech classification in real time during the playing of the game. Results and samples are registered in PLAY and can be consulted and analyzed a posteriori by the therapist as seen in Fig. 5. The UI of Fanima can be seen in Fig. 3 c).

The system architecture for the Fanima SG is composed of the child's mobile interface (the Fanima game), the therapist's game control interface, a web server and its database, both provided by the PLAY platform. PLAY provides the therapist with additional interfaces to analyse results and configure the game. Communication between Fanima and PLAY is achieved using an API, allowing user authentication, game structure and content retrieving, and game results and sample submission. PLAY provides a bidirectional channel for real-time interaction between the two interfaces, allowing to follow the current state of the exercise and interact with the patient session (e.g., receive the sampled speech and classify it, validating or assigning an error state).

The therapist's pervasive control interface is simple and

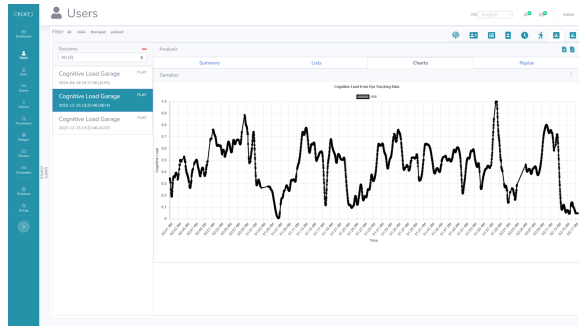


Fig. 4. Cognitive load signal recorded in PLAY.



Fig. 5. Captured audio sample from the Fanima SG.

straightforward with just a few controls, where the therapist can make the child repeat the level/chapter, choose the next chapter, and introduce some notes or issues detected in the child's performance. This way, the therapist can pervasively control the course of the session, manipulating the chapters and phonemes to work on, and the child feels total control and autonomy over the tasks to be performed. The PLAY platform supports the recording of each patient's sessions as well as level settings. The therapist can then follow the evolution of the sessions for each patient. Another positive point is that the results of each task can be accessed by other health professionals or clinics, depending on permissions. Therefore, patients may have a better follow-up in treatment across different therapists and clinics.

The repository of classified speech samples will provide the basis as datasets for training an ML model for autonomous classification as further work. Therefore, the system will have a component to evaluate the child's performance based on classification algorithms, replacing the therapist as the controller of the gameplay and letting the child to play without supervision. or giving additional support. A user study with 37 therapists allowed validation of the concept, theoretical background and platform integration. A user study with children is ongoing.

The Pikita SG is a VR application to tackle food phobia in children, focusing on speech and occupational therapy. The structure of the game is setup in PLAY using the SG design tools as seen in Fig. 6. The VR SG and PLAY establish WS connection for real-time bidirectional communication. When a patient starts the application, the configuration for the game session is requested from PLAY. Each patient may have a distinct combination of game parameters and these can also change in different sessions to reflect the current patient needs. The game parameters are configured

in the therapist interface and sent to Pikita. The configuration interface is displayed in Fig. 7. The game area dimensions can be adjusted. This is the area where the user can interact with the fear objects. Game warnings and safe area behavior can be defined including possibility of darkening the screen when approaching the safe limits and automatic activation of the safe environment. The fear and confort objects can also be setup. The game allows distinct phobia targets currently focusing on fruits (e.g., apple, orange). The confort objects are visualized when the user is in the safe environment and are represented by farm animals (e.g., chicken, pig). Finally, the score for each action can be assigned. After sending these set of parameters the game starts and the therapist interface changes to display the game flow control elements. This control interface allows the therapist to change the game scenario, activate specific game features, or stop the game. During the game session, samples for game status changes and patient interactions are sent to PLAY using the HTTP API. All data from each session is available for posterior analysis and is associated to the patient profile in the PLAY platform.

SGs integrated with PLAY use several sensor devices used for capturing signals and interaction samples including:

- IoT Wearable devices (e.g., Shimmer IMU, Bitalino, Unicorn Hybrid Black BCI)
- IoT VR devices (e.g., Oculus Quest II, Pico Neo3 Pro)
- other devices (e.g., Smartphone, Tablet)

The architecture of integration of SGs with IoT devices and sensors and PLAY is displayed in Fig. 8. IoT devices (e.g., sensors) connect to the SG application with a direct (e.g., integrated cable) or wireless link (e.g., Bluetooth, HTTP). A system component such as the lab streaming layer (LSL) can be used to capture data streams. Captured data is sent to the PLAY server using HTTP API requests. An example sample of ECG and EDA signals captured with the Bitalino sensor

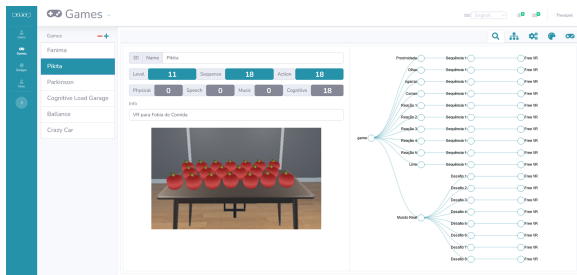


Fig. 6. The Pikita SG structure defined in PLAY.

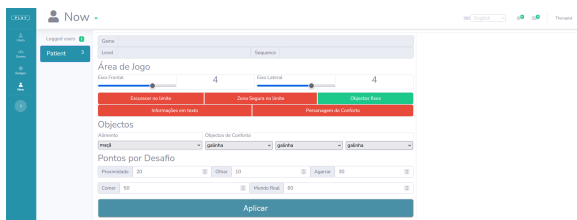


Fig. 7. The Pikita SG configuration interface.

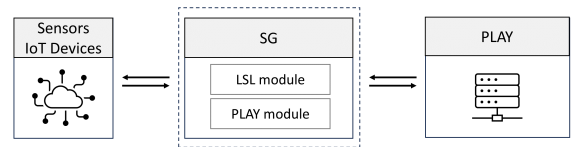


Fig. 8. Architecture of integration of SG with IoT sensors and PLAY.



Fig. 9. Sample of captured EDA and ECG signals using the Bitalino sensor.

using LSL can be seen in Fig. 9. PLAY also provides API calls to communicate with IoT actuator devices (e.g., smart gateway) to control external devices and provide tools to build more immersive experiences.

Some of these works were made in collaboration with the clinic Cresce Com Amor (CCA). Studies with therapists and patients were conducted to validate the proposed solutions.

V. CONCLUSIONS AND FUTURE WORK

The architecture of a DT for personalized SG-based therapy has been designed, and the main components have been identified.

Using the PLAY platform as the main component in the DT framework brings several advantages, such as the user profile, data repository, API, data visualization, SG design, and integration.

SG integration through the API is an essential feature of this system. The PLAY tools allow users to design the structure of each game, which is a relevant feature for personalization. The configuration tools allow users to adjust the game settings to the current needs of each patient.

Data captured from sensor interactions in each SG is used to build the user profile. The PLAY data repository maintains the historical information of each patient. PLAY provides tools for a digital shadow of each patient. The user profile directly relates to a patient digital shadow. Recorded patient profile changes, history, results, and samples compose the patient digital thread. Data from the repository is available, in real-time, for analysis or usage as source for AI/ML algorithms.

The DT of each patient will allow simulation and prediction. Therapists will be able to setup distinct therapy programs and apply them to the patient DT, obtaining the predicted outcomes. This will provide tools for deciding which therapy would be more suitable for a distinct patient. The final goal is to be able to provide a more personalized and adapted therapy program to a distinct patient considering the current constraints and needs. A meaningful global patient representation that can be used as input for ML algorithms is still being researched. There is still work to be done to achieve the functionality of the application of scenarios, simulation and prediction.

Although the research is on the right track to achieve the proposed goal of designing a DT for personalized SG-based therapy, future work includes continuous development and integration of new SG applications and sensors. The inclusion of more data sources will provide the DT algorithms with a richer dataset from which to learn the features of patients. The research will continue for the implementation and integration of AI and ML algorithms for the smart layer of the DT model based on the proposed framework.

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